

BHARGAV REDDY BATTU

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Full-Stack Software Engineer with experience building production-grade Agentic AI systems, scalable cloud infrastructure, and real-time web applications using AWS, Python, React and Java. Proven ability to deliver end-to-end solutions from RAG-based AI pipelines and adaptive backend architectures to responsive user-facing applications. Strong focus on combining AI, cloud-native services, and full-stack engineering to build impactful, enterprise-ready products from the ground up.

EDUCATION:

University of Florida, Gainesville, Florida

Master of Science in Computer Science

Jan 2024-Dec 2025

CGPA: 3.66/4

Vellore Institute of Technology, Vellore

Bachelors in Electronic and Communication

Aug 2019-Apr 2023

CGPA: 8.52/10

TECHNICAL SKILLS:

- **Programming** : Java 17(Core & Advance), Python, C#, Typescript.
- **Backend Technologies** : Spring Boot 3.x, Spring MVC, Spring Data JPA, Hibernate/JPA, Fast API, LangGraph, REST APIs.
- **AI&ML** : Agentic AI, Retrieval-Augmented Generation (RAG), AWS Bedrock, AWS AgentCore Memory, MediaPipe, Prompt Engineering.
- **Web & Data Base** : React.js, JavaScript, HTML5, CSS3, Bootstrap, MySQL, Supabase.
- **Cloud & DevOps** : AWS (EC2,ECS,ECR,S3,IAM,Amplifier,CloudWatch,SES), Docker, CI/CD Pipeline, Git.
- **Testing & CS Fundamentals** : Data Structures,Operating System, Computer Networks, SDLC.

PROFESSIONAL EXPERIENCE:

AALMV, Software Engineer

January-2025-Present|Tampa, Florida

- Engineered a production Agentic AI tutoring backend from scratch using Python, FastAPI, AWS Bedrock, boto3 and AWS AgentCore Memory, implementing a 5-node reasoning pipeline with RAG, per-student semantic memory, and mastery-driven teaching mode adaptation IAM-authenticated, Dockerized, and monitored via CloudWatch, improving response by 60%.
- Migrated a hardcoded Supabase Edge Function into a fully stateful LangGraph multi-node AI agent, replacing static rule-based responses with a dynamic, mastery-aware session orchestration system driven by real-time per-student performance data.
- Provisioned and maintained end-to-end AWS cloud infrastructure including EC2, IAM role-based access control, security groups, and AWS Amplify CI/CD pipelines, achieving automated zero-downtime deployments and 24/7 production uptime via systemd service automation.
- Developed a full-stack React and TypeScript web application from scratch, implementing multi-role JWT authentication across 3 user roles (student/educator/admin), real-time performance tracking, equation rendering, and interactive kinematic motion graphs.
- Architected a scalable Supabase PostgreSQL backend with normalized multi-table schema (alpha_chats, mastery_map, student_progress), row-level security (RLS) for per-student data isolation, real-time subscriptions, and persistent storage for assessment scores and per-topic mastery analytics across 9 AP Physics topics.

University of Florida, Software Engineer

June 2025 - December 2025|Gainesville, Florida

- Integrated MediaPipe 21-landmark hand tracking with Unity's Animation Rigging system in C#, implementing dual-hand, multi-layer dead-zone filtering, and inverse kinematics constraints (MultiPositionConstraint, ChainIKConstraint) for real-time finger articulation and bevel orientation detection, improving frame stability by 30% and reducing visualization latency by 20%.
- Developed a Python and .NET real-time data analytics pipeline capturing hand-tracking accuracy metrics including timestamps, attempt numbers, and error distances, applying statistical computation (mean error, standard deviation) and automating CSV report generation, eliminating 50% of manual validation tasks and improving data reliability by 35%.
- Built an interactive Unity debugging interface with landmark freeze controls enabling background inference, plus real-time bevel angle display, dual-hand visualization, and configurable UI state improved debugging efficiency by 25% and productivity by 40%.

Techurate Pvt Ltd, Software Engineer

January 2023 - December 2023|Bangalore, Karnataka

- Developed scalable backend systems using Spring Boot, Hibernate, MySQL, and microservices, leveraging query optimization and connection pooling to support 10K daily transactions, reduced API latency by 30% and content update latency by 50%.
- Built real-time analytics and reporting pipelines with multithreading, scheduled tasks, and data aggregation, integrating behavior-based recommendations into a React dashboard to improve processing efficiency by 30%, product discoverability by 20%, and maintain 95% data consistency under concurrent load.

PROJECTS:

• **Distributed Systems Simulation (Gossip & Push-Sum Algorithms)**

Developed a distributed system simulation using Gleam (OTP actor model) implementing Gossip and Push-Sum algorithms with asynchronous message passing, convergence detection, and performance tracking across multiple network topologies (full, line, 3D, imperfect), building a scalable and fault-tolerant architecture supporting 1000+ nodes with optimized execution efficiency.

• **Real-Time Chat Application (QuickChat)**

Developed a real-time chat application using Java socket programming and client-server architecture, supporting 100+ concurrent users with low-latency messaging, connection management, and protocol-level communication; improved throughput by 30% and delivery reliability by 25% via optimized packet handling and multithreading.